

Kenya Wines & Spirits POS — Business, Compliance & System Requirements Analysis

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Executive Summary

Tessera POS should be a Kenyan wines & spirits POS built around three things: an append-only stock ledger that controls shrinkage, eTIMS submission as a tracked state of every sale rather than a receipt printout, and M-PESA reconciliation matched to individual sales. Every Kenyan POS we reviewed already covers generic sales, stock, M-PESA and eTIMS. None of the sources describes controls specific to wines & spirits. That gap is the product opportunity.

Key conclusions

- **Competitive baseline:** eTIMS integration, M-PESA (STK Push and Till/Paybill), offline selling with sync, real-time stock and multi-branch reporting are table stakes. Four of the five vendor-reviewed sources advertise all of them.
- **Differentiation:**
 - bottle-size variants that cannot be mixed up
 - breakage and shrinkage tracking by reason, person and branch
 - blind cash-ups and blind stock counts
 - maker-checker approval for price changes, voids and refunds
 - retail and wholesale price tiers on one product
- **eTIMS:** KRA offers two system-to-system routes, OSCU (always online) and VSCU (not always online, bulk invoicing). Either one needs sandbox testing, a bio-data submission and KRA certification. For a POS that must keep selling offline, **VSCU is the likely fit** (REQUIRES VALIDATION against the v2.0 specifications).
- **Data protection:** customer data should be optional, and walk-in sales should need no personal data. ODPC registration depends on the business's turnover and headcount, so the business itself must confirm whether it has to register.
- **Alcohol regulation:** none of the supplied sources covers alcohol licensing, age limits, trading hours or excise stamps. Every alcohol-specific control here is marked **Additional regulatory verification required**.
- **Stack:** Laravel API, Next.js + Mantine frontend, PostgreSQL, mirroring NAMRIMS. The MVP is one business with multiple branches, retail sales and eTIMS. Wholesale credit, loyalty and e-commerce are deferred.

Evidence labels used throughout

Label	Meaning
SOURCE	Stated explicitly in one of the supplied sources (linked)
INFERRED	Reasonable deduction from the sources or from standard retail practice
RECOMMENDATION	Our design choice for Tessera POS
REQUIRES VALIDATION	Must be confirmed with KRA, ODPC, a county authority, a lawyer or the business owner before it is built as a rule

No code will be written until this analysis and the MVP scope in Phase 8 are approved.

Phase 1 — Source Analysis

Only two of the twelve sources are authoritative: the KRA pages and the ODPC FAQs. Four are vendor marketing about POS. Three (Pangoni, Silqu, Nyumba Zetu) are property-management software, not POS. They are useful only as reference patterns for M-PESA reconciliation and eTIMS invoicing.

Source	Type	What it explicitly establishes	Reliability
KRA — eTIMS system-to-system integration	Regulator	Defines OSCU ("invoicing is always online") and VSCU ("bulk invoicing and is not always online"). Both are API integrations. OSCU and VSCU spec documents v2.0 exist. Sandbox at etims-sbx.kra.go.ke. Integrate yourself or through a KRA-verified third-party integrator.	Authoritative

Source	Type	What it explicitly establishes	Reliability
KRA eTIMS portal — integrator certification	Regulator	Six steps: review specs → sandbox development → bio-data form + documents → KRA technical and administrative review → interim certificate → final review and full certification. Third-party integrators must show three qualified technical staff, tax compliance, a notarised solvency declaration, architecture documents and the software on read-only optical disk.	Authoritative
ODPC — FAQs	Regulator	All data controllers and processors must register. Exemption applies only when turnover is below KES 5M and staff are fewer than 10, and does not apply in listed high-risk sectors. Fees run from KES 4,000 to 40,000. Certificates last 24 months. Material changes must be notified within 14 days. "Retailers" is given as an example of a data controller.	Authoritative

Source	Type	What it explicitly establishes	Reliability
Manwa Advocates — data protection	Law firm commentary	Consent must be specific, documented and not bundled. Breach notification to ODPC within 72 hours. DPIA for high-risk processing. Cross-border transfer rules (s.48-50). Penalty up to KES 5M or 1% of turnover. Says one serving copy of personal data must stay on a server in Kenya.	Secondary. Its registration test differs from ODPC's, and its localisation statement is broader than we understand the regulation to be (REQUIRES VALIDATION)
Veira — POS system Kenya	Vendor marketing	Kenyan POS baseline: eTIMS receipt, M-PESA Till / Paybill / Pochi la Biashara + card, offline with sync, per-attendant login and audit trail, multi-branch dashboard. Plans KES 2,999-9,999 per month.	Vendor self-description
Veira — Best POS for Kenyan SMBs	Vendor comparison (written by a competitor)	Compares Veira, Tiwi, Pesapal Sabi, Uzapoint and SimbaPOS. Treats eTIMS, offline, M-PESA prompts and total hardware cost as the key selection criteria. Says eTIMS is "mandatory from January 2026".	Biased. The date claim REQUIRES VALIDATION with KRA

Source	Type	What it explicitly establishes	Reliability
Tiwi — POS systems in Kenya	Vendor marketing	eTIMS through OSCU/VSCU, STK Push, offline sync. Vertical features for supermarkets, restaurants and pharmacies (batch and expiry tracking). Hardware list: touch terminal, 80mm thermal printer, scanner, PDQ, cash drawer. No liquor vertical.	Vendor self-description
Exotic Digital Access — Best POS Kenya	Vendor marketing	eTIMS with QR code and buyer PIN capture for B2B, STK Push, 48-hour offline queue, low-stock and expiry alerts, unlimited branches. Market figures (80%+ of non-cash payments via M-PESA; 2-5% stock loss without real-time inventory; 392K businesses flagged; KES 1M penalty).	Unsourced statistics, not relied on . It calls OSCU a "physical device", which contradicts KRA
Uzapoint	Vendor site	Broadest suite: sales, inventory, procurement, AR/AP, P&L, multi-branch, audit trails (top tier only), loyalty, eTIMS, mobile money. Explicitly lists liquor stores and bars as industries served. No public pricing.	Vendor self-description

Source	Type	What it explicitly establishes	Reliability
Pangoni	Property management	M-PESA rent collection with automated reconciliation; eTIMS-aligned invoicing.	Not a POS; reconciliation pattern only
Silqu	Property management	M-PESA and bank auto-matching; role portals (caretaker, security).	Not a POS; reconciliation pattern only
Nyumba Zetu	Property management	eTIMS invoice transmission; M-PESA matching per invoice; double-entry ledger; QuickBooks export.	Not a POS; ledger and eTIMS pattern only

What the sources do not cover: alcohol licensing, legal drinking age, trading hours, excise duty, excise stamps, county permits, and the eTIMS field-level specification. The spec PDFs are referenced but were not part of the supplied material.

Kenyan POS Market Analysis

The Kenyan SMB POS market competes on compliance and payments, not on vertical depth. Every POS vendor in the sources leads with eTIMS, M-PESA and offline mode. Only Uzapoint names liquor stores, and none describes liquor-specific controls.

Common features (SOURCE, across Veira, Tiwi, EDA, Uzapoint)

- Checkout with barcode scanning and receipt printing
- eTIMS invoice transmission (EDA: QR code on receipt, buyer KRA PIN for B2B)
- M-PESA STK Push and Till/Paybill with automatic matching; card through a PDQ
- Offline selling with later sync (EDA advertises a 48-hour queue)
- Real-time stock, low-stock alerts, expiry tracking (EDA, Tiwi pharmacy)
- Multi-branch dashboards; per-attendant login and audit trail (Veira)
- Subscription pricing of roughly KES 1,000–10,000 per month per business, with hardware sold separately or bundled

Gaps for a wines & spirits business (INFERRED)

Gap	Why it matters in liquor retail
No size-variant discipline	The same brand at 250ml, 750ml and 1L sells at very different prices. A wrong-variant scan is both a revenue leak and a stock error.
Breakage is not a first-class event	Glass breaks during receiving, shelving and sale. The generic "adjustment" hides it inside other losses.
No blind counts or blind cash-ups	High-value, easily resold stock needs counts and cash-ups where staff do not see the expected figure first.
Weak maker-checker controls	Voids, refunds and price overrides are the usual cashier fraud routes. Sources mention audit trails but not approval workflows.
Retail and wholesale on one till	Wines & spirits shops often sell by the case to bars and restaurants. Generic tools split this into separate modules or tiers.
Case vs bottle units	Stock arrives in cases (e.g. 12 × 750ml) but sells by the bottle. Conversion errors inflate or deflate stock.
Excise-stamp and age controls absent	Not covered by any source. REQUIRES VALIDATION before deciding whether the POS should enforce them.

Features not worth copying (RECOMMENDATION)

- Payroll/HR, AI assistants, e-commerce storefronts and WhatsApp marketing (EDA, Uzapoint). They add scope without solving liquor-shop problems.
- Restaurant table management, unless the business also runs a bar counter (open question).
- Full double-entry accounting. Export to the accountant's system instead.

Phase 2 — Wines & Spirits Business Requirements

The core product rule is simple: a **sellable item is a specific pack size of a specific product**, never just a brand. Each one has its own barcode, SKU, prices and stock balance (RECOMMENDATION).

Business profile the system must scale across (from the brief)

- Single shop: 1 till, owner and 1–2 cashiers
- Multi-branch: 2–10 outlets, optional central warehouse, branch managers, storekeepers, an accountant
- Customers: walk-in retail (most sales), plus wholesale buyers (bars, restaurants, events) and corporate accounts
- Typical categories: wine (red, white, rosé, sparkling/champagne), whisky, vodka, gin, brandy, rum, tequila, liqueurs, beer and cider, mixers, soft drinks, water, ice, accessories (glasses, openers, gift bags). Each business turns on only the categories it sells.

Product structure

flowchart TD

```

B[Brand<br/>Johnnie Walker] --> P[Product<br/>JW Black Label]
P --> V1[Variant 200ml<br/>SKU JWB-200]
P --> V2[Variant 375ml<br/>SKU JWB-375]
P --> V3[Variant 750ml<br/>SKU JWB-750]
P --> V4[Variant 1L<br/>SKU JWB-1000]
V3 --> PK[Pack: Case of 12<br/>own barcode]
```

A product holds shared attributes. Each variant is the stock-keeping, sellable unit. A pack (such as a case of 12) is a purchasing and wholesale unit that converts to a fixed number of variants.

Product fields

Level	Fields
Brand	Name, producer, country of origin
Product	Name, brand, category, subcategory (e.g. Whisky → Blended Scotch), ABV %, description, image, active flag
Variant (sellable)	Volume in ml, container (bottle/can/keg), SKU, one or more barcodes, eTIMS item classification code and tax type (REQUIRES VALIDATION of code lists), cost price (weighted average, calculated by the system), retail price, wholesale price, minimum price floor, reorder level and reorder quantity per branch, batch/expiry tracking flag, active flag
Pack	Pack name (case/carton/crate), units per pack, pack barcode, pack wholesale price

Level	Fields
Supplier link	Preferred suppliers per variant, supplier item code, last purchase cost

How variants are kept apart (RECOMMENDATION)

1. The POS always shows the volume as part of the item name: "JW Black Label **750ml**". The display name is generated from product + volume, so it cannot be typed inconsistently.
2. Barcode uniqueness is enforced across all variants and packs. Scanning a case barcode adds a case of 12, not one bottle.
3. Product search groups results by product, with variants listed as separate lines showing volume and price.
4. A 1L bottle cannot be priced below its 750ml sibling without a warning (a soft price-ladder check).
5. Stock counts are always entered per variant, never per product.

Pricing (RECOMMENDATION)

- Retail and wholesale prices are held per variant, per branch, with dated price lists so history is never overwritten.
- Wholesale pricing applies only to customers flagged as wholesale, or to case quantities.
- A price below the floor, or a manual price override, requires manager approval.
- Batch and expiry tracking is optional per variant. It is useful for beer, cider, RTDs and mixers, rarely needed for spirits (INFERRED).

Sales Workflow

A sale is finished only when payment is confirmed, stock is deducted and the eTIMS invoice is **submitted and acknowledged**. Printing a receipt alone does not make a sale compliant (RECOMMENDATION, consistent with KRA's description of eTIMS as system-to-system transmission).

```

stateDiagram-v2
    [*] --> Open: cashier starts sale
    Open --> Suspended: park sale
    Suspended --> Open: recall
    Open --> Voided: void (approval)
    Open --> AwaitingPayment: tender
    AwaitingPayment --> Paid: all tenders confirmed
    AwaitingPayment --> Open: payment failed
    Paid --> Completed: stock posted + receipt
    Completed --> EtimesPending: queued
    EtimesPending --> EtimesSigned: KRA ack
    EtimesPending --> EtimesFailed: error
    EtimesFailed --> EtimesPending: retry

```

Once a sale is Completed, its line items and totals are **immutable**. Any later change is a separate document, such as a return (credit note) or a correction.

Standard sale steps

1. The cashier is logged in with an open shift and an opening float.
2. Scan or search the item. The line shows name + volume + unit price (retail by default).
3. Change the quantity, or scan a case barcode to add a pack.
4. Discounts: line or basket level, within the role's limit. Above the limit needs a manager PIN, and the reason is recorded.
5. Customer: optional for retail. Required for wholesale, credit and B2B invoices, where the buyer's KRA PIN goes on the eTIMS invoice (SOURCE: EDA mentions buyer PIN capture; field rules REQUIRE VALIDATION).
6. Tender: one or more payment methods until the balance is zero.
7. Complete: the sale is committed in one database transaction (sale + lines + payments + stock ledger entries).
8. The eTIMS job is queued. The receipt prints with the eTIMS details once signed, or marked "eTIMS pending" if offline (REQUIRES VALIDATION whether a pending receipt is acceptable under VSCU rules).
9. Dashboards and reports update from the posted records.

Alternative scenarios

Scenario	Handling	Approval
Cash	Enter tendered amount; change calculated; drawer opens	—

Scenario	Handling	Approval
M-PESA STK Push	Enter phone → prompt → callback confirms → tender marked paid	—
M-PESA Till/Paybill (customer-initiated)	Cashier picks the matching unallocated confirmation from a live list	—
Card	Amount keyed on a standalone PDQ; cashier records the approval code and last 4 digits	—
Bank transfer	Mainly for wholesale; recorded as pending until the accountant confirms receipt	Accountant confirms
Split payment	Any combination; each tender stored separately	—
Credit sale	Only for approved account customers within their credit limit; creates a receivable	Manager, if over limit
Partial payment	Allowed only on credit/wholesale invoices; the rest becomes a receivable	Per credit policy
Suspended sale	Parked with the cashier's ID; recallable at the same branch; auto-expires at shift close	—
Line void (before tender)	Removes a line; logged	Manager PIN for high-value items (RECOMMENDATION)
Void whole sale (before completion)	Cancels an unpaid sale; logged with reason	Manager PIN
Cancel after completion	Not allowed. Must be a return, creating an eTIMS credit note	Manager
Refund / return	See Scenario 3 in User Journeys	Manager
Receipt reprint	Allowed, printed as "COPY" with a reprint count; logged	—

Inventory Workflow

Stock on hand is never edited directly. It is the running sum of an **append-only stock ledger**: every receipt, sale, transfer, breakage and count produces a signed movement row with a reason, user and source document (RECOMMENDATION). This is the main defence against shrinkage.

```

flowchart LR
    S[Supplier] --> PO[Purchase order]
    PO --> GRN[Goods received<br/>+ verification]
    GRN --> L[(Stock ledger)]
    L --> W[Warehouse / branch]
    W --> T[Transfer]
    T --> W
    W --> POS[POS sale]
    POS --> L
    W --> ADJ[Breakage / loss / count]
    ADJ --> L
  
```

Every arrow into the ledger is a document with a number, a user, a timestamp and an approval state.

Movement types

Movement	Effect	Who creates	Who approves	Notes
Goods received (GRN)	+	Storekeeper	Branch manager (if variance vs PO)	Counts by case, converts to bottles; records damaged-on-arrival separately
Sale	–	Cashier (automatic)	—	Posted at sale completion
Customer return (restockable)	+	Cashier	Manager	Only sealed, undamaged bottles go back to sellable stock
Customer return (non-restockable)	0 to sellable, + to quarantine	Cashier	Manager	Goes to a quarantine location

Movement	Effect	Who creates	Who approves	Notes
Transfer out / in	– source, + destination	Storekeeper	Manager at each end	Shown as in transit between dispatch and receipt
Breakage	–	Storekeeper / cashier	Manager	Reason and photo optional; attributed to stage (receiving, shelf, sale, transit)
Expired / spoiled	–	Storekeeper	Manager	Batch-tracked variants only
Missing / theft	–	System (from count variance)	Manager + owner above threshold	Never keyed directly as free text
Return to supplier	–	Storekeeper	Manager	Linked to a debit/credit note from the supplier
Stock count variance	±	System	Manager (owner above threshold)	Result of a blind count
Opening balance	+	Admin	Owner	Once, at go-live; locked afterwards

Stock counts (RECOMMENDATION)

- **Blind counts:** counters enter quantities without seeing system stock. Variance is calculated only after submission.
- **Cycle counts:** a daily count of the top 20 high-value variants (spirits over a set price) plus a full count each month.
- A count freezes a snapshot of the expected quantity. Sales during the count are handled by comparing against the ledger at the moment the count was submitted.
- Variances above the threshold (e.g. KES 5,000 or 2 bottles of premium spirits) need a recount before they can be posted.

Valuation and profit

- Cost method: **weighted average cost** per variant per branch, recalculated on each GRN (RECOMMENDATION; FIFO is more complex and rarely needed for non-perishables). The accountant must confirm this method (REQUIRES VALIDATION).

- COGS = quantity sold × weighted average cost at the time of sale, stored on the sale line so historical margin never changes.
- Gross profit = net sales (excluding VAT) – COGS.
- Shrinkage % = (breakage + missing + expired value) ÷ COGS for the period, reported per branch, variant and reason.

Low-stock and reorder

- Reorder level and quantity are set per variant per branch.
- A nightly job plus real-time checks raise low-stock alerts, which feed a suggested purchase order or a suggested transfer from the warehouse.

Purchasing & Supplier Management

Purchasing uses three-way matching. The purchase order (PO), goods received note (GRN) and supplier invoice must agree before the invoice becomes payable. This stops paying for bottles that never arrived (RECOMMENDATION).

```
flowchart LR
    PO[PO<br/>manager approves] --> GRN[GRN<br/>storekeeper counts]
    GRN --> INV[Supplier invoice<br/>accountant enters]
    INV --> M{3-way match}
    M -->|ok| AP[Supplier balance]
    M -->|variance| Q[Query supplier]
    AP --> PAY[Supplier payment]
```

A GRN posts stock immediately. The invoice and payment only affect the supplier balance.

Supplier features

Feature	MVP?	Notes
Supplier register	Yes	Name, KRA PIN, contacts, payment terms, bank/M-PESA details. Company data, so ODPC exposure is minimal (INFERRED)
Purchase orders	Yes	Per branch or warehouse, in cases or bottles; statuses Draft → Approved → Sent → Partially received → Closed
Goods received notes	Yes	Against a PO or standalone (manager approval); records quantity received, quantity damaged on arrival, batch/expiry

Feature	MVP?	Notes
Supplier invoices	Yes	Invoice number, date, amounts, VAT; linked to GRNs. Whether purchase invoices must be pulled from or confirmed in eTIMS REQUIRES VALIDATION against the spec
Supplier payments	Phase 2	Recorded (cash/bank/M-PESA B2B); partial payments allowed
Supplier balances and statements	Phase 2	Aged payables 0–30, 31–60, 61–90, 90+ days
Returns to supplier	Yes	Stock out, then awaiting the supplier's credit note
Purchase history	Yes	Per supplier and per variant; last cost, cost trend
Supplier performance	Future	Fill rate (received ÷ ordered), lead time, damage rate on arrival

Customer Management

Most wines & spirits sales are anonymous walk-in sales, so the default customer is "Walk-in" and no personal data is collected. Customer records exist only where the business needs them: wholesale pricing, credit, and B2B eTIMS invoices that need the buyer's PIN.

Feature	Needed?	Verdict	Reason
Walk-in customer	Yes	MVP	Default for every sale; no data captured
Registered wholesale customer	Yes	MVP	Wholesale price tier; business name + KRA PIN for eTIMS
Corporate / B2B customer	Yes	MVP (same record as wholesale)	Buyer PIN on the invoice so the buyer can claim input VAT (INFERRED; REQUIRES VALIDATION of fields)

Feature	Needed?	Verdict	Reason
Credit accounts + limits	Sometimes	Phase 2	Common with bars and restaurants, but a source of bad debt; needs statements and aging
Customer invoices / statements	With credit	Phase 2	Tied to credit accounts
Purchase history	Yes, for registered customers	MVP	Falls out of the sales records at no extra cost
Promotions (price rules)	Yes	Phase 2	e.g. "buy 6 wine, 10% off"; time-boxed, approved by the owner
Loyalty points	Doubtful	Future / optional	Needs phone numbers and consent; value unproven for liquor (INFERRED). Also check whether alcohol promotion rules restrict loyalty (REQUIRES VALIDATION)
Marketing SMS / WhatsApp	No	Excluded	Needs separate opt-in consent; outside the POS's job
Individual retail customer profiles	No, by default	Optional	Only with the customer's consent (e.g. for delivery)

Payment Architecture

Every payment is stored as its own tender record, with method, amount, external reference and a confirmation status. A sale is paid only when all its tenders are confirmed. M-PESA and bank tenders are confirmed by the provider's callback, not by the cashier's word (RECOMMENDATION).

```
sequenceDiagram
    participant T as Till (Next.js)
    participant A as Laravel API
    participant D as Daraja (M-PESA)
    T->>A: STK request (phone, amount, sale id)
    A->>D: STK Push
    D-->>Customer: PIN prompt
    D->>A: Callback (result, receipt no.)
    A->>A: Match to tender, mark confirmed
    A-->>T: Push update (paid)
```

If the callback does not arrive, the API queries the transaction status before the cashier can retry. This prevents a double charge.

Method	Integration	External provider?	Reconciliation
Cash	None; cash drawer opened through the printer port	No	Shift cash-up: expected vs counted, blind
M-PESA STK Push	Safaricom Daraja API (M-PESA Express)	Yes: Daraja credentials, shortcode, public HTTPS callback URL	Automatic, by receipt number
M-PESA Till / Paybill (customer-initiated)	Daraja C2B confirmation/validation URLs	Yes	Confirmations land in an "unallocated" pool; cashier or system matches them to a sale; end-of-day list of unmatched items
Pochi la Biashara	Mentioned by Veira (SOURCE). API availability REQUIRES VALIDATION	Unclear	Manual reference entry if there is no API
Card	Standalone PDQ from the acquiring bank; semi-integrated later	Bank/acquirer (Phase 2 integration)	Cashier keys the approval code; daily match against the settlement report
Bank transfer	None in MVP	No	Accountant confirms against the bank statement

Method	Integration	External provider?	Reconciliation
Credit (on account)	Internal	No	Receivables ledger

Cash drawer and shift control (RECOMMENDATION)

- Each cashier opens a shift with a counted float; one open shift per cashier per till.
- Cash drops (moving excess cash to the safe) are recorded during the shift with a manager witness.
- Closing: blind count by denomination → system shows expected vs counted → variance with reason → manager signs off.

The Daraja credentials, shortcodes and the eTIMS device are per business (and possibly per branch). Whether each branch gets its own Till is an open question.

Phase 3 — eTIMS Integration

Tessera POS should integrate with eTIMS system-to-system through a dedicated integration layer, most likely using **VSCU**, and must be certified by KRA before going live. Until each sale has a KRA-acknowledged invoice record, its status stays "pending", and failures are shown on a compliance screen.

What KRA explicitly states (SOURCE: [integration page](#), [certification page](#))

- eTIMS supports system-to-system integration between a taxpayer's invoicing/ERP system and KRA through an API.
- **OSCU** is "suitable for taxpayers whose invoicing is always online".
- **VSCU** is "suitable for taxpayers undertaking bulk invoicing and is not always online".
- KRA publishes OSCU and VSCU specification documents (v2.0), a sandbox registration guide and a sandbox at etims-sbx.kra.go.ke.
- A taxpayer can integrate itself or use a KRA-verified third-party integrator.
- Certification steps: review the specs → build and test in the sandbox → submit the bio-data form and documents → KRA technical and administrative assessment → **interim approval certificate** → final review → full certification.
- Third-party integrators must provide incorporation documents, CR12, a business permit, director IDs, tax compliance certificates, proof of three qualified technical staff, a notarised solvency declaration, architecture documents and the software on read-only optical disk.

Choosing OSCU or VSCU

Factor	OSCU	VSCU
KRA positioning	Always-online invoicing	Not always online; bulk invoicing
Fit for a shop with unreliable internet	Poor: sales that need a live signature stall when offline (INFERRED)	Better: designed for intermittent connectivity
Deployment	API calls to KRA	Per the spec, a KRA-supplied component runs within the taxpayer's environment (REQUIRES VALIDATION from the VSCU v2.0 spec)
Recommendation	Fallback option	Preferred, subject to spec review

(EDA's claim that OSCU is a "physical device" contradicts KRA's description of both as system-to-system integrations, so it is disregarded.)

Integration architecture (RECOMMENDATION)

```

flowchart LR
    POS[POS sale / return] --> OB[(eTIMS outbox table)]
    OB --> W[Queue worker<br/>Laravel Horizon]
    W --> ADP[eTIMS adapter<br/>OSCU or VSCU driver]
    ADP --> KRA[KRA eTIMS]
    KRA --> ADP
    ADP --> RES[(eTIMS responses)]
    RES --> RCPT[Receipt + reports]

```

The sale is committed together with an outbox row in the same database transaction, so a sale can never exist without its eTIMS job (the transactional outbox pattern). The adapter is behind an interface, so switching between OSCU and VSCU, or to a third-party integrator, does not touch sales code.

Invoice lifecycle to support

Event	eTIMS document	Notes
Normal sale	Sales invoice	Items with classification codes and tax types; buyer PIN optional for retail, needed for B2B

Event	eTIMS document	Notes
Customer return / refund	Credit note that references the original invoice	Cancelling an invoice after completion is not modelled; returns go through a credit note (REQUIRES VALIDATION of how the spec models this)
Price correction upward	Debit note	Rare; Phase 2
Receipt reprint	Copy receipt	Spec may define receipt types such as normal, copy, training and proforma (REQUIRES VALIDATION)
Items master	Item registration	Each variant may need registering with KRA item codes before it can be sold (REQUIRES VALIDATION)
Purchases and stock	Purchase / stock movement reporting	The OSCU/VSCU specs are understood to include purchase and stock-movement endpoints. If so, the stock ledger must map to them (REQUIRES VALIDATION)

What we store per submission: our invoice number, KRA-returned invoice/receipt number, control/signature data, QR code payload, SCU/device ID, request and response bodies, attempt count, status (pending / signed / failed / rejected), timestamps. Exact field names come from the spec (REQUIRES VALIDATION).

Error handling, retries and duplicates

- Idempotency: each invoice is sent with the same POS invoice number on every retry. Before resending after a timeout, the adapter checks whether KRA already accepted it.
- Retry with exponential back-off (1 min → 2 → 4 → ... capped at 30 min). Validation errors (e.g. a bad item code) are **not** retried; they go to a "needs fixing" queue for the admin.
- Invoices are submitted strictly in sequence per device if the spec requires sequential numbering (REQUIRES VALIDATION).
- Dashboard alerts: pending older than 1 hour, any rejection, a failed-queue backlog.
- Daily reconciliation: count and value of POS sales vs eTIMS-acknowledged invoices per branch and day; any mismatch is flagged.

Offline: sales continue and invoices queue in the outbox. Whether a receipt issued before KRA acknowledgement is compliant, and how long invoices may stay queued, depend on the VSCU rules (REQUIRES VALIDATION). The system must not claim eTIMS compliance for queued invoices.

Certification path for Tessera: SoftClans as a third-party integrator versus each client self-integrating is a business decision, see Open Questions. Either way, sandbox development can start once the v2.0 spec PDFs are obtained.

Data Protection / ODPC

The POS should collect the least personal data that works: none for walk-in sales, and business details for wholesale and B2B. Each business using Tessera is a **data controller**. If Tessera is hosted by SoftClans, SoftClans is a **data processor** and needs a processing agreement with each client (SOURCE: ODPC defines controller and processor and lists retailers and cloud providers as examples).

Registration (SOURCE: [ODPC FAQs](#))

- Controllers and processors must register with ODPC. A business is exempt only if turnover is below KES 5M **and** it has fewer than 10 employees, and not in a listed high-risk sector.
- Fees: KES 4,000 (micro/small), 16,000 (medium), 40,000 (large). Certificates last 24 months. Material changes must be notified within 14 days.
- Manwa Advocates states the test differently (turnover above KES 5M *or* more than 10 staff). ODPC's own wording governs, and each client should confirm its status (REQUIRES VALIDATION).

Personal data the POS touches

Data	Whose	Collected when	Lawful basis (INFERRED)	Minimisation rule
Name, phone	Wholesale/credit customer contact	Account opening	Contract	Business name preferred over personal name
KRA PIN	B2B buyer	B2B invoice	Legal obligation (tax invoice)	Only when the buyer asks for a PIN invoice

Data	Whose	Collected when	Lawful basis (INFERRED)	Minimisation rule
Phone number	M-PESA payer	STK Push	Contract (payment)	Stored masked on the sale (e.g. 0712***678); full number only in payment logs, access-restricted
M-PESA name from callback	Payer	Payment	Contract	Not shown on receipts
Card data	Card payer	Never	—	POS stores only approval code + last 4 digits; no card numbers ever
ID number / date of birth	Customer	Not collected by default	—	Age checks are visual; see Alcohol section
Staff name, phone, login, activity logs	Employees	Onboarding	Contract / legitimate interest	No national ID stored in the POS unless the owner requires it
Supplier contacts	Supplier staff	Registration	Contract	Business contacts only

Controls (RECOMMENDATION, informed by [Manwa Advocates](#))

- **Access control:** role-based; customer phone numbers are visible only to managers and above; exports need the Owner/Admin permission and are logged.
- **Audit trail:** every view of a customer's full record and every export is logged (who, when, what).
- **Consent:** only for optional uses (loyalty, marketing, delivery). Stored per purpose, with date and channel, never pre-ticked (Manwa: consent must be specific, documented and not bundled).
- **Data subject requests:** an admin screen to export one customer's data, correct it, or anonymise it. Anonymisation replaces personal fields but keeps the financial records, because tax records must be retained (retention period **REQUIRES VALIDATION** with KRA and the accountant).

- **Retention:** configurable. Default proposal: inactive customer personal data anonymised after 24 months of inactivity; financial documents kept for the statutory period.
- **Security:** TLS everywhere, encrypted backups, hashed passwords, MFA for Owner/Admin, session timeout on tills.
- **Breach response:** incident log and procedure. Manwa states ODPC must be notified within 72 hours (REQUIRES VALIDATION against the Act and regulations).
- **Hosting location:** Manwa states that a serving copy of personal data must stay on a server in Kenya. Our understanding is that this applies to specific categories, not all processing (REQUIRES VALIDATION with counsel before choosing a non-Kenyan cloud region).

Alcohol-Industry Considerations

None of the supplied sources covers alcohol regulation, so **every item below requires additional regulatory verification**. The POS should offer these as **configurable business-policy controls**, not hard-coded legal rules, until a lawyer or the county licensing office confirms them.

Area	What the POS could do	Enforce in POS or keep as policy?	Status
Liquor licence (county)	Store licence number, type, expiry per branch; alert 30/60 days before expiry; optionally print the licence number on receipts	POS reminder; compliance stays with the business	Additional regulatory verification required: licence types and whether receipts must show them
Business / single business permit	Store permit number and expiry per branch	Reminder only	Additional regulatory verification required
Minimum age	Optional "age confirmed" prompt on the first alcohol line of a sale (yes/no, no ID captured)	Configurable prompt; the check itself is a staff duty	Additional regulatory verification required: the legal age and whether a record of the check is expected

Area	What the POS could do	Enforce in POS or keep as policy?	Status
Permitted trading hours	Per-branch sell window; alcohol items blocked outside it, with a manager override that is logged	Configurable, off by default	Additional regulatory verification required: hours vary by county and licence type
Sale to visibly intoxicated persons	None; staff judgement	Business policy	Additional regulatory verification required
Excise stamps / product authenticity	Optional capture of stamp scan on receiving; flag products received without stamps	Phase 2 / Future	Additional regulatory verification required: how retailers verify stamps and whether POS integration exists
Sales records for inspection	Complete immutable sales and stock records, exportable per branch and date range	Built in (ledger design)	Additional regulatory verification required on the specific records inspectors expect
Product traceability	Supplier + GRN + batch per unit received; trace a batch to its sales	Built in for batch-tracked variants	Additional regulatory verification required
Regulatory reporting	Configurable exports (CSV/PDF)	Built in as generic exports	Additional regulatory verification required: whether any periodic returns exist

Principle: the POS supplies evidence (records, logs, reminders). Legal compliance stays with the licensed business.

RBAC & Segregation of Duties

Roles are bundles of granular permissions, scoped to one or more branches. No single front-line role can both create and approve a stock loss, a refund or a price change (RECOMMENDATION). The roles below are defaults. The owner can clone and adjust them, but the four segregation rules at the end are enforced in code.

Action	Owner	Admin	Branch Mgr	Cashier	Storekeeper	Accountant
Sell	✓	✓	✓	✓	—	—
Discount within limit (e.g. 5%)	✓	✓	✓	✓	—	—
Discount above limit / price override	✓	✓	Approve	Request	—	—
Void line / sale before completion	✓	✓	Approve	Request	—	—
Refund / return	✓	✓	Approve	Request	—	—
Receive stock (GRN)	✓	✓	✓	—	Create	View
Transfer stock	✓	✓	Approve	—	Create/dispatch/receive	View
Record breakage / adjustment	✓	✓	Approve	Report breakage	Create	View
Stock count	✓	✓	Approve variance	—	Count (blind)	View
View cost & profit	✓	✓	Own branch	—	—	✓
Financial reports	✓	✓	Own branch	—	—	✓
Manage products	✓	✓	Request	—	—	—
Configure prices	✓	Create	Request	—	—	—
Manage suppliers / invoices	✓	✓	View	—	View	✓
Manage users & roles	✓	✓ (not Owner)	—	—	—	—
View audit logs	✓	✓	Own branch	—	—	✓
End-of-day reconciliation	✓	✓	Approve	Blind count	—	Review
eTIMS settings / failed queue	✓	✓	View	—	—	View

Enforced segregation rules

1. The requester and approver of a void, refund, price override or stock adjustment must be different users. Approval is by the manager's own PIN or login, never a shared code.

2. Price changes are made by Admin and take effect after Owner approval (or immediately if the Owner makes them). Both are logged.
3. Cashiers never see expected cash before counting. Storekeepers never see expected stock before a blind count.
4. Nobody, including the Owner, can delete a completed sale, payment or stock movement. Corrections are reversing documents.

Dashboard

The owner's dashboard answers four questions: how much did we sell, where did the money go, are we losing stock, and are we compliant. Ten KPIs are enough. Each one must lead to an action.

KPI	Why it matters	Who sees it
Net sales today (vs same day last week)	Headline performance; the weekly comparison removes weekday effects	Owner, Mgr
Sales by tender (cash / M-PESA / card / credit)	Tells cash-handling exposure and what reconciliation to expect	Owner, Mgr, Accountant
Gross profit & margin %	Liquor margins are thin; volume alone misleads	Owner, Accountant
Discounts, voids, refunds (value + count, by cashier)	Main cashier-fraud signals	Owner, Mgr
Shrinkage this month (breakage + missing, value and % of COGS)	The key wines & spirits loss metric	Owner, Mgr
Cash variance by shift	Early warning of cash theft or errors	Owner, Mgr
Low-stock variants (count, top 10 list)	Lost sales from stockouts of fast movers	Mgr, Storekeeper
Top 10 and slow-moving variants (no sale in 60 days)	Buying decisions; slow stock ties up cash	Owner, Mgr
Stock value at cost	Capital tied up; basis for insurance	Owner, Accountant
eTIMS status (pending / failed / oldest pending age)	Compliance risk; must be close to zero	Owner, Admin

KPI	Why it matters	Who sees it
Branch comparison (multi-branch only)	Branch-level sales, margin and shrinkage	Owner
Receivables & payables (Phase 2)	Cash-flow exposure from credit customers and suppliers	Owner, Accountant

Left out on purpose: transaction heat maps, basket-size charts and customer counts. They are interesting but rarely acted on in a liquor shop (INFERRED). They can be added as reports later.

Reports

All reports run off the immutable ledgers. They can be filtered by branch, date range, category, brand and user, and exported to CSV/PDF. Exports are permission-gated and logged.

Group	Report	MVP
Sales	Daily / weekly / monthly sales summary (gross, discounts, returns, net, VAT)	Yes
Sales	Sales by variant, product, category, brand	Yes
Sales	Sales by cashier, by branch, by tender	Yes
Sales	Voids, discounts & price overrides register (with approver)	Yes
Sales	Returns & refunds register	Yes
Sales	Hourly sales pattern	Phase 2
Inventory	Current stock by branch (qty + value at cost)	Yes
Inventory	Stock movement ledger (per variant, per document)	Yes
Inventory	Stock valuation as at a date	Yes
Inventory	Adjustments & losses by reason (breakage, missing, expired, damaged on arrival)	Yes
Inventory	Stock count variance report	Yes
Inventory	Low stock / reorder suggestions	Yes

Group	Report	MVP
Inventory	Transfers (in transit, received, discrepancies)	Yes
Inventory	Slow-moving & dead stock	Phase 2
Inventory	Expiry report (batch-tracked items)	Phase 2
Financial	Revenue, COGS, gross profit by period / branch / category	Yes
Financial	Shift cash-up and variance report	Yes
Financial	M-PESA reconciliation (matched / unmatched)	Yes
Financial	Purchases by supplier	Yes
Financial	Receivables & payables aging	Phase 2
Financial	Expenses (petty cash)	Phase 2
Compliance	eTIMS transaction status (signed / pending / failed)	Yes
Compliance	Failed / rejected eTIMS submissions with errors	Yes
Compliance	POS vs eTIMS daily reconciliation	Yes
Compliance	Audit trail (filter by user, action, entity)	Yes
Compliance	Licence & permit expiry list	Phase 2

Phase 6 — User Journeys

Prices are illustrative KES amounts, VAT-inclusive at 16%. Product names are examples only.

Scenario 1 — Normal retail sale (2 wine + 1 whisky, cash)

1. Cashier Aisha has an open shift at Westlands with a KES 5,000 float.
2. She scans "4th Street Sweet Red **750ml**" twice → $2 \times 1,050 = 2,100$. She scans "JW Black Label **750ml**" → 4,800. Total **6,900**, of which VAT 951.72.
3. The optional age prompt appears on the first alcohol line (if enabled). She taps "Confirmed".
4. Tender: cash 7,000 → change 100. The drawer opens.
5. The system commits in one transaction: sale #WL-000482 (Completed), 2 lines with unit cost captured, a cash tender, stock ledger –2 wine and –1 whisky at Westlands, and an eTIMS outbox row.

6. The worker submits to eTIMS. KRA returns the invoice number and signature/QR data, and the status becomes Signed.
7. The receipt prints with KRA invoice details and QR. The dashboard's net sales for today rise by 5,948.28 (excluding VAT).

Scenario 2 — M-PESA sale

1. Customer buys Gordon's Gin 750ml at 2,300. Cashier picks M-PESA and enters 0712 345 678.
2. The API calls Daraja STK Push. The till shows "Waiting for customer..." with a 60s countdown.
3. The customer enters their PIN. Daraja calls back with success and receipt number e.g. SIP4XK9ABC.
4. The API matches the callback to the tender. The tender is Confirmed, the sale Completed, stock is deducted and eTIMS is queued.
5. If there is no callback within 60s, the API queries the transaction status. If it is still unknown, the cashier can switch to "customer paid by Till" and pick the matching confirmation from the unallocated list. **The cashier cannot mark M-PESA as paid without a confirmation.**
6. End of day: the M-PESA reconciliation report shows every tender matched to a Daraja receipt, plus any unallocated confirmations to investigate.

Scenario 3 — Customer return

Two days later a customer returns an unopened bottle of 4th Street 750ml from sale #WL-000482.

1. The cashier looks up the original sale by receipt number or QR scan and selects 1 unit to return.
2. The system checks that the item was on that sale, has not already been returned, and is within the return window (policy, e.g. 7 days).
3. The cashier records the condition: sealed → restockable. The manager approves with their PIN.
4. Effects:
 - **Stock:** +1 to Westlands sellable stock at the original unit cost.
 - **Sales:** a return document of -1,050, reported as returns (not by editing the original sale).
 - **Payment:** refund of 1,050 in cash from the cashier's drawer (or an M-PESA B2C reversal in Phase 2).
 - **Customer:** if the sale had a registered customer, their history shows the return.

- **eTIMS:** a credit note referencing the original KRA invoice is queued and signed.

5. An opened or broken bottle goes to quarantine as damaged, not back to sellable stock.

Scenario 4 — Broken bottle

1. Storekeeper Brian drops a Jameson 1L while shelving at Westlands.
2. He records a Breakage: variant Jameson 1L, qty 1, stage "shelving", optional photo and note.
3. Stock does not move yet. The document is Pending approval.
4. The branch manager approves. The ledger posts -1 at weighted average cost (e.g. KES 3,150), reason Breakage, and the adjustment is linked to Brian and the manager.
5. The shrinkage KPI and the "losses by reason" report update. Repeat breakages by one person or stage show up in the monthly review.

Scenario 5 — Low stock

1. Smirnoff Vodka 750ml at Westlands has a reorder level of 24 and a reorder quantity of 48.
2. A sale takes on-hand stock from 25 to 23. The system raises a low-stock alert.
3. The alert appears on the dashboard and in the manager's notifications. The low-stock report lists it with 30-day sales velocity and days of cover.
4. The manager chooses to request a transfer from the warehouse (which holds 120), or to generate a draft PO to the supplier for 4 cases (48).

Scenario 6 — Branch transfer

flowchart LR

```
R[Branch B requests<br/>24 × Smirnoff 750ml] --> A[Branch A
manager<br/>approves]
```

```
A --> D[Storekeeper A<br/>dispatches 24]
```

```
D --> IT[In transit<br/>-24 at A]
```

```
IT --> RC[Storekeeper B<br/>receives 23 + 1 broken]
```

```
RC --> S[+23 at B<br/>1 breakage 'transit']
```

The transfer document records quantity requested, dispatched and received. The one-bottle shortfall becomes a breakage with stage "transit", which needs manager B's approval. Stock is never counted twice or lost in between: it sits in an **in-transit** location until received.

Scenario 7 — End-of-day cash-up

1. Aisha closes her shift and counts cash by denomination **without seeing the expected figure** (blind count). She enters KES 41,300.
2. The system calculates:

$$\text{Expected} = \text{Opening float} + \text{Cash sales} - \text{Cash refunds} - \text{Cash drops}$$

For example: 5,000 + 58,450 – 1,050 – 21,000 = **41,400**.

3. Variance: 41,300 – 41,400 = **-100 (short)**. Aisha enters a reason. The branch manager recounts and signs off.
4. M-PESA and card totals show as confirmed-by-provider figures. There is nothing to count, but unmatched items are listed.
5. The shift report is locked. Variances accumulate per cashier on the dashboard.

Phase 4 — Competitor Gap Analysis

Tessera must match the Kenyan baseline in every row and differentiate in the last four rows. Evidence below is only what the sources state. "Not stated" means the source is silent, not that the feature is missing.

Capability	Competitor evidence (sources)	Tessera requirement
POS sales	Veira , Tiwi , EDA , Uzapoint : checkout, barcode, receipts	Parity + case/bottle scanning and volume-in-name display
Inventory	All four: real-time stock, low-stock alerts; Tiwi/EDA: expiry and batch	Append-only ledger, reasoned movements, blind counts, in-transit stock
Multi-branch	Veira dashboard; EDA "unlimited branches"; Uzapoint top tier only; Tiwi enterprise plan	Built in from day 1; branch-scoped roles
M-PESA	Veira: Till/Paybill/Pochi + auto-reconciliation; Tiwi/EDA: STK Push	STK Push + C2B pool + callback-only confirmation + daily reconciliation
eTIMS	Veira "certified"; Tiwi via OSCU/VSCU; EDA: QR + buyer PIN; Uzapoint: stated; Pesapal via partner	Certified OSCU/VSCU adapter, outbox, credit notes, POS-vs-eTIMS reconciliation, failure queue

Capability	Competitor evidence (sources)	Tessera requirement
Supplier management	EDA: supplier management; Uzapoint: procurement, AP	PO → GRN → invoice three-way match; damaged-on-arrival capture
Reporting	Veira/EDA dashboards; Uzapoint P&L, consolidated	Liquor-specific shrinkage and loss-by-reason reports
RBAC	Veira: per-attendant login; Silqu: role portals (non-POS)	Granular permissions + enforced maker-checker
Audit trail	Veira: audit trails; Uzapoint: top tier only	Every tier; before/after values; immutable
Wines & spirits controls	Uzapoint lists liquor stores and bars as industries; no source describes liquor-specific controls	Variant discipline, breakage workflow, blind cash-up/count, optional age prompt and trading-hour rules, licence expiry reminders

Differentiation opportunities: shrinkage visibility, maker-checker approvals, retail and wholesale tiers on one till, eTIMS reconciliation that shows the gap rather than just claiming compliance.

Not to copy: payroll/HR, AI assistants, e-commerce storefront, WhatsApp marketing, restaurant table management, free hardware bundles (a commercial choice, not a product feature).

Phase 5 — Proposed System Modules

The suggested structure is mostly right, with three changes. **Payments** merges into Sales and Shifts. **Compliance** (eTIMS + licences) becomes a module of its own. **Expenses** is deferred.

#	Module	Scope	Phase
1	Dashboard	KPIs listed above, per role	MVP
2	POS (till)	Sell, tenders, suspend/recall, returns, reprint, shift open/close, cash drops	MVP
3	Sales & Shifts (back office)	Sales list, returns register, approvals queue, cash-ups, M-PESA reconciliation	MVP

#	Module	Scope	Phase
4	Catalogue	Brands, products, variants, packs, barcodes, categories, price lists	MVP
5	Inventory	Stock on hand, ledger, adjustments, breakages, counts, transfers, reorder	MVP
6	Purchasing	Suppliers, POs, GRNs, supplier invoices, returns to supplier	MVP (payments and balances in Phase 2)
7	Customers	Walk-in default, wholesale/B2B records; credit accounts later	MVP lite
8	Compliance	eTIMS config, outbox monitor, failed queue, reconciliation; licence register	MVP (licences Phase 2)
9	Reports	Sales, inventory, financial, compliance catalogues	MVP
10	Organisation	Business profile, branches, locations (shop floor, store, quarantine, in transit), tills, printers	MVP
11	Users & Roles	Users, roles, permissions, branch scopes, PINs, MFA	MVP
12	Audit Log	Searchable, read-only	MVP
13	Settings	Tax, receipt layout, policies (discount limits, return window, age prompt, trading hours)	MVP
—	Expenses / petty cash	Phase 2	Phase 2
—	Promotions, credit accounts, supplier payments	Phase 2	Phase 2

```

flowchart TD
    D[Dashboard] --> POS[POS till]
    D --> SS[Sales & Shifts]
    D --> INV[Inventory]
    D --> PUR[Purchasing]
    D --> CMP[Compliance]
    D --> ADM[Admin: Org, Users, Settings, Audit]

```

Catalogue, Customers and Reports sit under Admin and Inventory in the navigation. This keeps the top level to six items for small shops.

Phase 7 — Technical Architecture

Tessera uses the NAMRIMS stack and repo layout: a Laravel API, a Next.js + Mantine web app, PostgreSQL, Redis queues and Docker Compose for dev and prod. The till runs as a PWA with local storage so it can keep selling offline (RECOMMENDATION).

```

flowchart LR
    TILL[Till PWA<br/>Next.js + Mantine<br/>IndexedDB] -->|HTTPS/JSON| API[Laravel API<br/>Sanctum auth]
    BO[Back office<br/>Next.js + Mantine] --> API
    API --> PG[(PostgreSQL)]
    API --> RD[(Redis<br/>queues, cache)]
    RD --> WK[Horizon workers]
    WK --> ET[eTIMS adapter<br/>OSCU/VSCU]
    WK --> MP[Daraja M-PESA]
    MP -->|callbacks| API
    API --> WS[Reverb websockets]
    WS --> TILL

```

Callbacks from Safaricom need a public HTTPS endpoint. The eTIMS adapter's placement (in the cloud, or on-premise per branch if VSCU requires it) depends on the spec (REQUIRES VALIDATION).

Layer	Choice	Reason
Repo	Monorepo backend/ , frontend/ , docker/ , docs/ , scripts/ (same as NAMRIMS MIS)	Same team conventions and CI
Backend	Laravel (current LTS-supported release), PHP 8.3+	NAMRIMS parity

Layer	Choice	Reason
Auth	Sanctum SPA/token auth; till PIN login on top of a device token; MFA for Owner/Admin	Fast cashier switching without weak shared passwords
Permissions	spatie/laravel-permission with branch-scoped role assignment	Mature and granular
Queues	Redis + Horizon	eTIMS outbox, M-PESA status checks, alerts
Realtime	Laravel Reverb	Push M-PESA confirmations to the till
Frontend	Next.js (App Router) + Mantine + TanStack Query	NAMRIMS parity; Mantine has strong data-table and form components
Offline	PWA service worker + IndexedDB (Dexie) on the till only	Keep selling offline; the back office can require a connection
Database	PostgreSQL 16+	Transactions, row locks, partial indexes, JSONB for eTIMS payloads
Money	Integers in cents (bigint) or <code>numeric(14,2)</code> ; never floats	Rounding safety
Printing	ESC/POS 80mm via a small local print bridge or WebUSB; browser print as fallback	Thermal printers are the norm (SOURCE: Tiwi hardware list)
Hosting	Cloud VM or Kenyan data centre with Docker Compose; region depends on the ODPC localisation check	See Data Protection

API style: REST under `/api/v1` , grouped by module (`/pos/sales` , `/inventory/movements` , `/purchasing/grns` , `/compliance/etims/outbox`). Every write accepts an `Idempotency-Key` header, which the offline sync depends on.

Database Architecture

Financial and stock tables are **insert-only**. Balances are derived or cached projections, and PostgreSQL triggers block UPDATE/DELETE on posted rows. History cannot be silently overwritten (RECOMMENDATION).

Domain	Tables	Notes
Organisation	businesses, branches, locations (floor, store, quarantine, in_transit), tills, devices	One business per deployment in MVP; business_id on every table for future multi-tenancy
Identity	users, roles, permissions, role_user_branch, user_pins, sessions	Branch-scoped roles
Catalogue	brands, categories, products, variants, variant_barcodes, packs, price_lists, prices (effective_from/to), tax_codes, etims_item_codes	Prices are dated rows, never edited in place
Parties	customers, suppliers, customer_consents	Customer personal fields are nullable; anonymisation flag
Sales	sales, sale_lines (unit_price, discount, tax, unit_cost snapshot), returns, return_lines, suspended_sales	Completed sales are immutable
Payments	tenders (method, amount, reference, status), mpesa_transactions (raw callbacks), mpesa_allocations	Raw provider payloads kept for reconciliation
Shifts	shifts, cash_movements (float, drop, payout), cash_counts (by denomination)	Blind count stored before expected is shown
Inventory	stock_movements (the ledger: variant, location, qty $\pm$, unit_cost, type, document_type/id, user, approved_by, reason), stock_balances (projection), stock_adjustments, stock_counts, stock_count_lines, transfers, transfer_lines, batches	stock_balances can always be rebuilt from stock_movements
Purchasing	purchase_orders, po_lines, grns, grn_lines, supplier_invoices, supplier_returns, supplier_payments (Phase 2)	GRN posts stock movements

Domain	Tables	Notes
eTIMS	<code>etims_outbox</code> (document ref, payload, status, attempts, next_attempt_at), <code>etims_submissions</code> (request, response, KRA invoice no., signature, QR, SCU id), <code>etims_config</code>	One outbox row per sale, return or credit note
Approvals	<code>approval_requests</code> (type, requester, approver, status, reason)	Maker-checker for voids, refunds, overrides, adjustments
Audit	<code>audit_logs</code> (actor, action, entity, entity_id, before JSONB, after JSONB, reason, branch, ip, device, at)	Append-only; partitioned by month
Offline	<code>sync_batches</code> , <code>idempotency_keys</code>	Deduplicates offline uploads

Key relationships: product 1→n variants 1→n barcodes; sale 1→n lines, 1→n tenders, 1→n stock_movements, 1→1 etims_outbox; return → original sale; transfer → two movement pairs (out, in-transit, in).

Concurrency: posting a sale locks the relevant `stock_balances` rows (`SELECT ... FOR UPDATE`) in a fixed order to avoid deadlocks. Negative stock is allowed but flagged (configurable), so the till never blocks a customer because of a counting error.

Numbering: document numbers are gapless per branch and type (e.g. WL-S-000482), allocated from a sequence table inside the transaction. Offline tills use a device prefix (WL-T2-000031).

Offline Architecture

When the internet drops, the till keeps selling for cash (and card on a standalone PDQ). M-PESA falls back to manual reference entry, with the reference verified on reconnect. Everything queues locally and syncs idempotently. Offline POS operation and eTIMS compliance are separate problems: the POS can keep trading offline, but eTIMS compliance of those sales depends on KRA's VSCU/OSCU rules (REQUIRES VALIDATION).

Question	Design answer
Can sales continue?	Yes. The till has cached catalogue, prices, barcodes, stock snapshot, open shift and user PINs (hashed) in IndexedDB.

Question	Design answer
How are payments handled?	Cash: normal. Card: standalone PDQ, approval code keyed in. M-PESA STK: unavailable. Till/Paybill: cashier enters the customer's M-PESA code from the SMS, and the sale is marked "unverified tender" until matched to Daraja C2B data on reconnect. Unmatched after sync → manager alert. Credit sales: blocked offline (credit limit cannot be checked).
How is eTIMS handled?	Sales are stored with status "eTIMS pending" and uploaded to the server outbox on reconnect. If VSCU runs locally at the branch (REQUIRES VALIDATION), invoices can be signed on the LAN even without internet.
How are duplicates prevented?	Each offline sale gets a client-generated UUID and a device-prefixed number. The server enforces uniqueness on UUID via <code>idempotency_keys</code> . Re-uploading returns the original result. eTIMS submissions use the same invoice number on every retry.
How does sync work?	Push queue first (sales, returns, cash movements in creation order), then pull (catalogue, price and stock deltas since the last sync cursor). Sync runs automatically on reconnect and every 60s while online.
How are conflicts resolved?	Sales are facts, so they are never rejected for price or stock. Server prices apply only to new sales. Offline sales keep the price charged; a price mismatch is flagged in a report. Negative stock from concurrent offline sales is flagged for a count. Master data (products, prices) is server-authoritative and read-only on the till.
Limits	Configurable maximum offline duration (e.g. 24h) and maximum offline sale value, after which the till warns and then requires manager override. The value should align with whatever limit the eTIMS spec sets (REQUIRES VALIDATION).
Unsupported offline	Returns against sales not on the device, stock transfers, GRNs, approvals needing a remote manager (a local manager PIN works)

Security & Audit Trail

Every sensitive action writes an immutable audit record with who, what, when, where, before, after, reason and reference. Audit records are written in the same database transaction as the action itself, so an action cannot happen without its log.

Audit record fields: actor user id and role · approver (if any) · action code · entity type + id · before (JSONB) · after (JSONB) · reason (required for the events below) · document reference · branch, till, device, IP · server timestamp + client timestamp (offline).

Event	Reason required	Approval	Extra
Price change	Yes	Owner	Old/new price per variant per branch; effective date
Discount above limit / override	Yes	Manager	Line, % and value
Void (line or sale)	Yes	Manager	Items and value voided
Refund / return	Yes	Manager	Original sale; restockable flag
Stock adjustment / breakage	Yes	Manager	Stage; photo optional
Stock transfer	On discrepancy	Manager each end	Requested / dispatched / received quantities
Stock count posting	For variances over threshold	Manager / Owner	Blind count values
User / role change	Yes	Owner or Admin	Permission diff
Product change	No	—	Field diff
eTIMS failure / manual resend	Automatic	—	Error code, attempts
Transaction correction	Yes	Manager + Owner	Always via a reversing document
Data export / customer record view	No	Permission-gated	Rows exported

Event	Reason required	Approval	Extra
Login, failed login, PIN reset	—	—	Lockout after 5 failures

Security baseline (RECOMMENDATION)

- TLS 1.2+; HSTS; secrets in env/Vault, never in the repo; Daraja and eTIMS credentials encrypted at rest.
- Passwords with Argon2/bcrypt; MFA (TOTP) for Owner/Admin; till PINs of 4–6 digits, valid only on registered till devices.
- Session timeout on the back office (30 min) and auto-lock on the till (2 min idle).
- PostgreSQL: app user has no UPDATE/DELETE on ledger and audit tables; triggers enforce this as well.
- Nightly encrypted backups with 30-day retention; a restore test each month.
- Laravel rate limiting on auth and callback routes; callback source validation for Daraja.
- OWASP ASVS L1 checklist before go-live; dependency scanning in CI (as in NAMRIMS workflows).

Phase 8 — MVP vs Future Scope

The MVP is what one multi-branch wines & spirits business needs to trade legally and control stock and cash on day one. Anything that adds a new customer type, integration or revenue stream waits.

Tier	Features
MVP	Till with cash, M-PESA STK + Till/Paybill matching, card (manual reference), split tender, suspend/recall, reprint · returns with credit notes · shifts with blind cash-up · catalogue with variants, packs, barcodes, dated retail/wholesale prices · append-only stock ledger, GRN, transfers with in-transit, breakage/adjustments with approval, blind counts, reorder alerts · suppliers, POs, GRNs, supplier invoices, returns to supplier · walk-in + wholesale/B2B customers with PIN · eTIMS adapter (sandbox → certification), outbox, failure queue, daily reconciliation · RBAC with maker-checker · audit log · MVP reports and dashboard · offline till (cash/card) · multi-branch, single business

Tier	Features
Phase 2	Credit accounts, limits, statements, aging · supplier payments and balances · promotions engine · debit notes · M-PESA B2C refunds · semi-integrated card terminals · expenses/petty cash · licence & permit register with reminders · trading-hour rules · expiry reports · slow-moving analysis · accounting export (QuickBooks/Xero/CSV) · hourly sales
Future / optional	Loyalty (subject to consent and alcohol-promotion checks) · excise stamp capture (subject to regulatory confirmation) · multi-tenant SaaS · native mobile owner app · delivery orders · bar/counter mode · supplier performance scoring · demand forecasting
Excluded	Payroll/HR · e-commerce storefront · marketing SMS/WhatsApp · AI assistants · full general-ledger accounting

Open Questions / Items Requiring Confirmation

KRA / eTIMS

- ☐ Obtain the OSCU and VSCU v2.0 specifications. Confirm which one suits intermittent-connectivity retail, and where VSCU runs (cloud or on-premise).
- ☐ Is a receipt issued before KRA acknowledgement acceptable? What is the maximum delay?
- ☐ Required invoice fields, receipt types, QR/signature content, numbering rules, credit/debit note references.
- ☐ Is item registration and stock/purchase reporting to eTIMS mandatory under system-to-system integration?
- ☐ Is the "mandatory from January 2026" claim (Veira) accurate, and what applies to existing businesses?
- ☐ Will SoftClans certify as a **third-party integrator** (three technical staff, solvency declaration, etc.), or will each client self-integrate?

ODPC / data protection

- ☐ Does each target client need to register (turnover and staff test)? Does SoftClans need to register as a processor?

- ☐ Does the Kenya server-copy requirement apply to POS customer data? This decides the hosting region.
- ☐ Statutory retention periods for sales and tax records.

Alcohol regulation (Additional regulatory verification required for all)

- ☐ Legal minimum age and whether retailers must record age checks.
- ☐ County trading hours per licence type for each target county.
- ☐ Whether receipts must show a licence number.
- ☐ Excise stamp verification duties for retailers.
- ☐ Any periodic sales or stock returns to county or national regulators.

Business owner decisions

- ☐ Pilot client: how many branches, is there a central warehouse, what share of sales is wholesale?
- ☐ Does any branch serve drinks on-premise (bar counter)? This changes scope.
- ☐ One M-PESA Till per branch, or one Paybill with account numbers?
- ☐ Discount limits, return window, breakage and variance approval thresholds.
- ☐ Costing method (weighted average proposed): confirm with the accountant.
- ☐ Hardware per till: touch terminal or PC, 80mm printer, scanner, cash drawer, PDQ.
- ☐ Single-business deployment first, or multi-tenant SaaS from day 1?

Recommended Next Steps

1. **Review and approve** this analysis and the MVP scope. Comment directly on any section.
2. **Get the eTIMS specs:** download the OSCU/VSCU v2.0 PDFs and register for the sandbox at etims-sbx.kra.go.ke. This is the longest-lead item.
3. **Legal checks:** send the alcohol and ODPC open questions to counsel and the pilot client's county licensing contact.
4. **Pilot client workshop:** confirm branches, wholesale share, thresholds and hardware.
5. **After approval:** scaffold the TESSERA-POS monorepo mirroring NAMRIMS (`backend/` Laravel, `frontend/` Next.js + Mantine, `docker/` with PostgreSQL + Redis, CI), then build in this order: catalogue → stock ledger → till and shifts → M-PESA → purchasing → eTIMS adapter against the sandbox → reports → offline till.

Sources consulted (opened 25 Sep 2026): [KRA eTIMS system-to-system](#) · [KRA eTIMS integrator certification](#) · [ODPC FAQs](#) · [Manwa Advocates](#) · [Veira — POS Kenya](#) · [Veira — Best POS 2026](#) · [Tiwi](#) · [Exotic Digital Access](#) · [Uzapoint](#) · [Pangoni](#) · [Silqu](#) · [Nyumba Zetu](#)